

PARTS LIST

Semiconductors:

- IC1 - CD4060 oscillator-cum-divider
 T1, T2 - BC547 npn transistor
 LED1 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 100-kilo-ohm
 R2, R6 - 10-kilo-ohm
 R3 - 150-kilo-ohm
 R4 - 1-mega-ohm
 R5 - 150-ohm
 VR1 - 2.2-mega-ohm preset

Capacitors:

- C1 - 100 μ F, 25V electrolytic
 C2 - 220nF ceramic disk

Miscellaneous:

- CON1 - 2-pin connector
 BATT.1 - 9V battery/PP3 type
 S1 - Push-to-on switch
 PZ1 - Piezo-buzzer

sistor T1 is connected to a 'gimmick' potential divider comprising sensitivity control preset VR1 (2.2-mega-ohm), fixed resistor R1 and the moisture sensor probe. In standby state (soil in wet condition), IC1 is disabled because its reset pin 12 is directly tied to +9 V through resistor R2. However, when the soil is dry, transistor T1 is forward biased by VR1 and R1. This

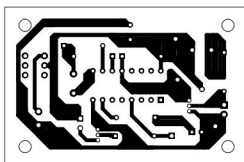


Fig. 3: An actual-size, single-side PCB for moisture monitor

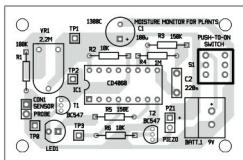


Fig. 4: Component layout for the PCB

enables IC1 to oscillate. As a result, LED1 starts blinking and piezobuzzer PZ1 sounds to indicate that the plant

needs water.

Moisture monitor requires a 9V supply. A PP3-size battery is recommended for compactness.

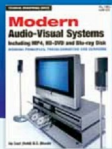
Construction and testing

An actual-size, single-side PCB for flora caretaker is shown in Fig. 3 and its component layout in Fig. 4. After construction, enclose the whole circuit in a suitable crystalline tube or box. The moisture sensor probe can be constructed using two small metal rods (steel or copper) mounted vertically with a small gap (about 10 mm) in between them. These two metal rods must be supported by a small insulated (wooden or plastic) board. You can place one flora caretaker in each plant pot for constant monitoring of soil moisture. ●

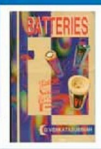
The author is a freelance writer and regular contributor to EFY. He has many articles published to his credit in India and abroad

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